

DOCUMENTATION

Node

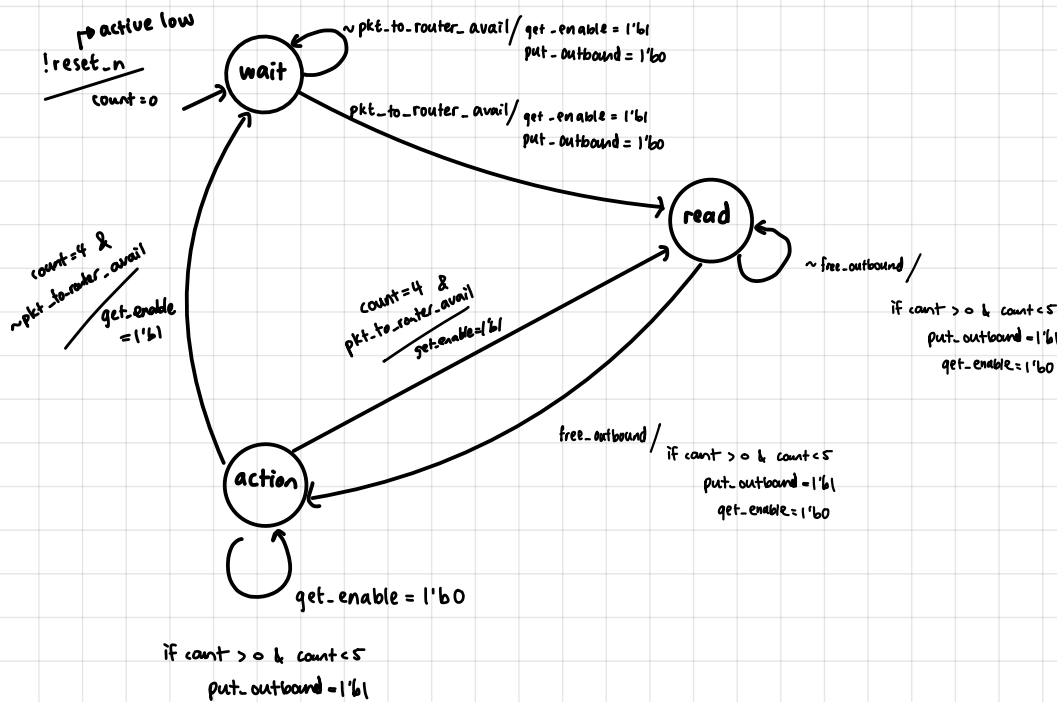
FSM (node to router)

↳ controls the packet communication between node & router

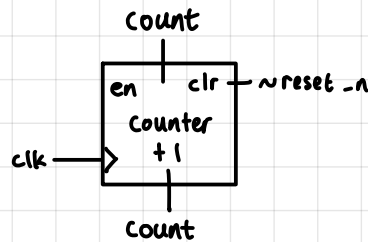
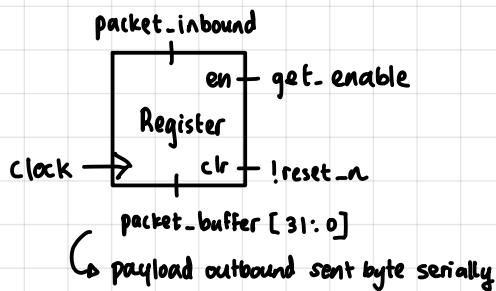
input: pkt_to_router_avail (when the queue in node is not empty)
free_outbound (when router is free)

output: put_outbound (status signal where node is transferring to router)
get_enable (read packet from queue)

FSM

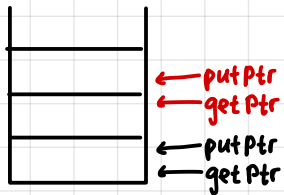


Datapath (node to Router)



always, combo
payload_outbound
 = packet_buffer [31:24] 1
[23:16] 2
[15:8] 3
[7:0] 4

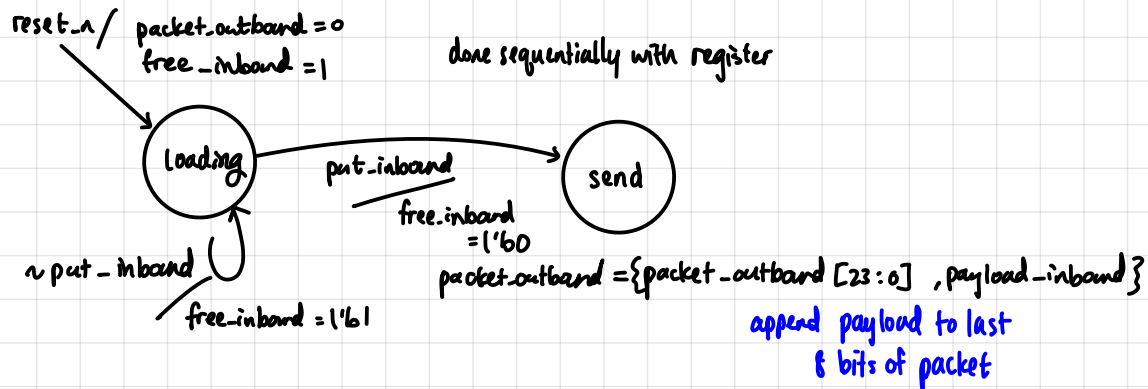
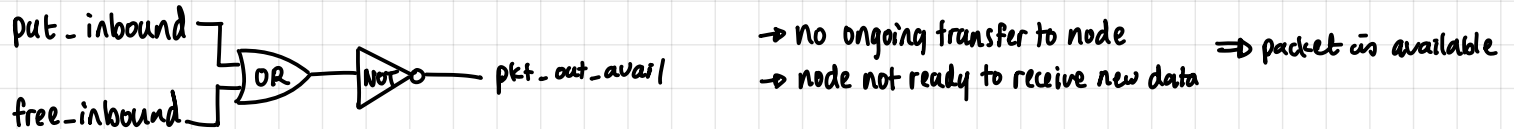
FIFO Implementation



(i.e increment or decrement)

- ↳ manipulate put & get pointer for inserting & removing from queue
- ↳ max slots of 4
- ↳ one ptr overflow, it wraps around to the first slot again at 0, 1, 2, 3 with the limitations of the # of bits or mod function

RouterToNode

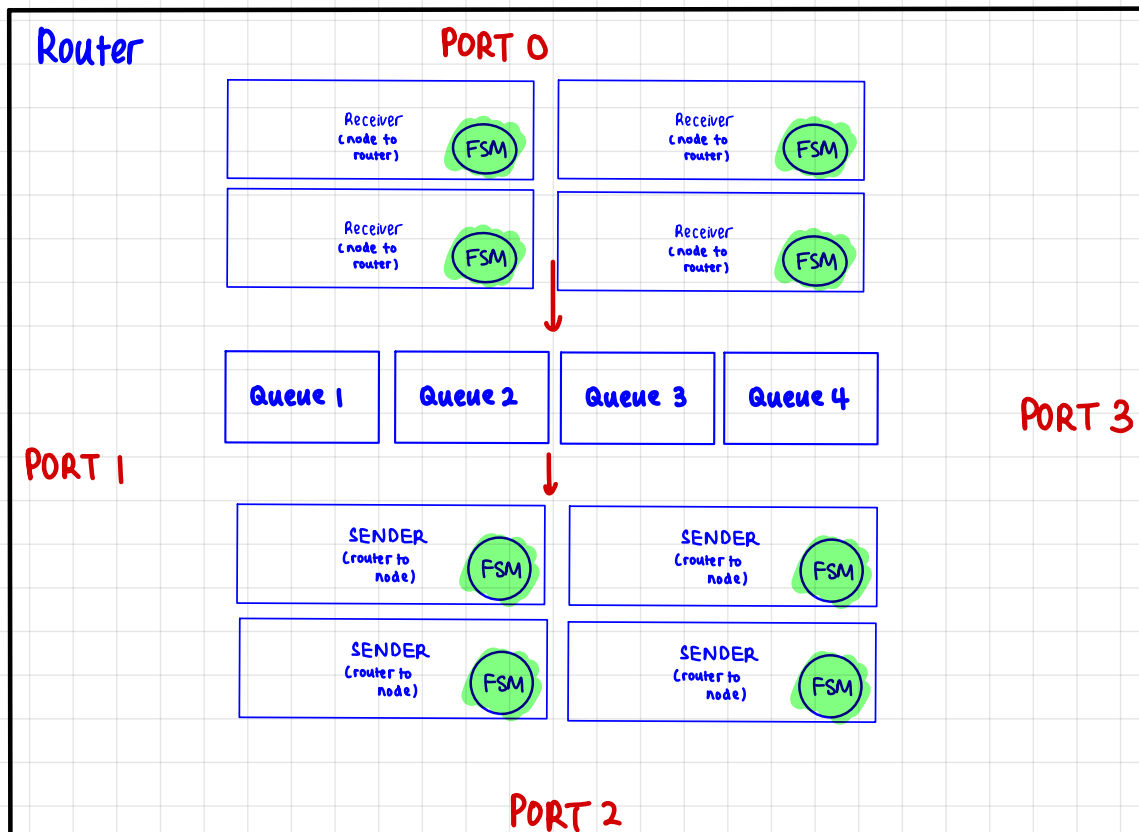


Router

- ★ 4 internal queues based on destination port of packet to ensure fairness among multiple packets ★

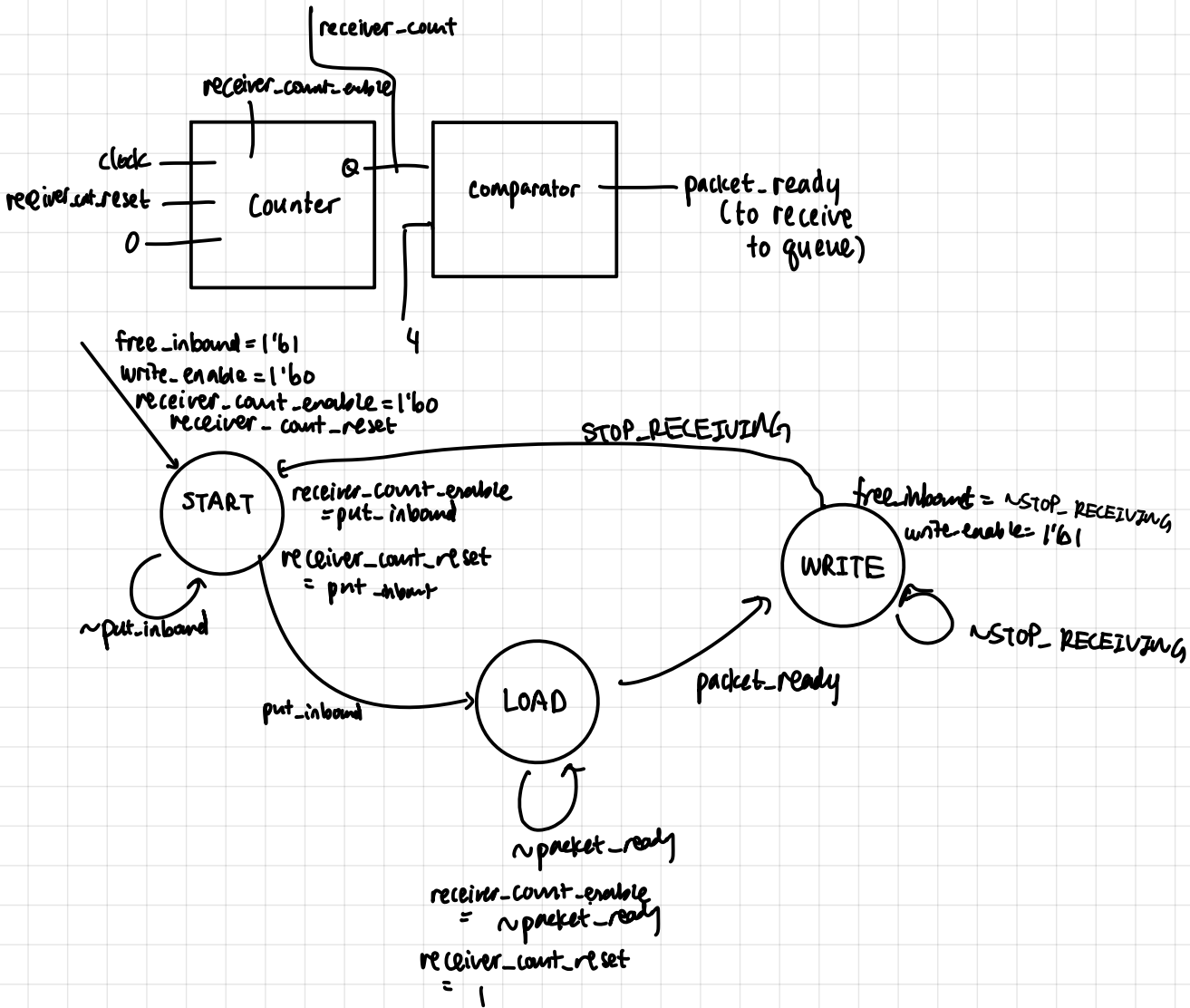
- ★ 4 receiver & 4 senders with FSM controlling the packets traffic ★

HIGH LEVEL OVERVIEW

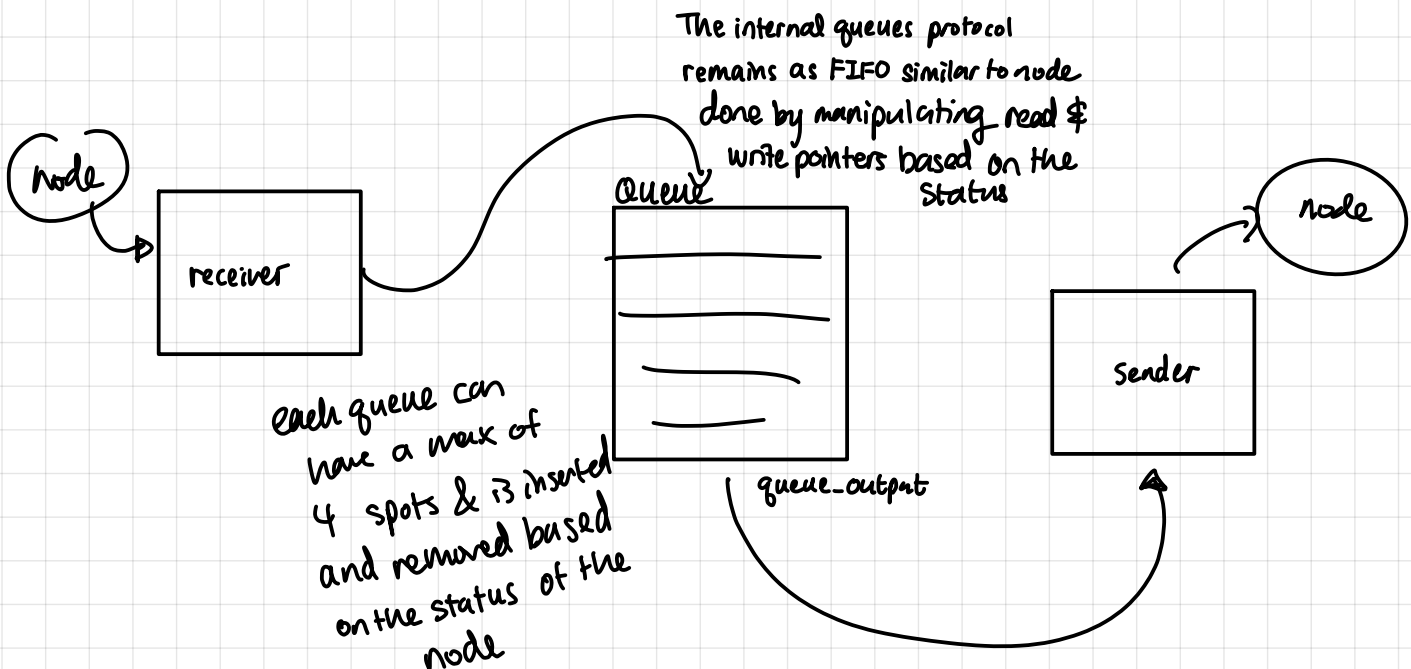


Receiver

→ receiver receives incoming packets & adds to its respective queue based on the packet dest. port



Internal Queue



Sender

